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Light-emitting device and light source module including the same

Abstract

A light source module is provided. The light source module includes a printed circuit board; a light-emitting device mounted on the printed circuit board, the light-emitting device including a plurality of cell blocks, each of which includes a plurality of light-emitting cells; and a plurality of controllers mounted on the printed circuit board, each of which is configured to drive a cell block corresponding thereto, from among the plurality of cell blocks. The plurality of cell blocks are electrically isolated from each other, the plurality of cell blocks include a first cell block and a second cell block, and a number of first light-emitting cells included in the first cell block is less than a number of second light-emitting cells included in the second cell block.

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Background/Summary

CROSS-REFERENCE TO RELATED THE APPLICATION

(1) This application claims priority from Korean Patent Application No. 10-2020-0063276, filed on May 26, 2020, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

(2) Apparatuses and methods consistent with example embodiments relate to a light-emitting device and a light source module including the same, and more particularly, to a light-emitting device including a plurality of blocks including a plurality of light-emitting cells, and a light source module including the light-emitting device.

2. Description of the Related Art

(3) Semiconductor light-emitting devices include elements such as light-emitting diodes (LEDs), and are increasingly being used as light sources because the semiconductor light-emitting devices have many advantages such as low power consumption, high brightness, and long lifespans. In recent years, there has been a growing interest in semiconductor light-emitting devices as units capable of replacing the existing halogen or xenon lamps used as light sources for headlamps or tail lamps for vehicles.

(4) When a semiconductor light-emitting device is applied as a light source to a lighting apparatus, it may be necessary to adjust brightness, a light orientation angle, and a light irradiation angle of the semiconductor light-emitting device to desired values. In particular, a headlamp or a tail lamp for a vehicle requires a semiconductor light-emitting device capable of adjusting the brightness of light according to surrounding conditions.

SUMMARY

(5) One or more example embodiments provide a light-emitting device, which has improved optical characteristics and reliability, and a light source module including the light-emitting device.

(6) According to an aspect of an example embodiment, a light source module includes a printed circuit board; a light-emitting device mounted on the printed circuit board, the light-emitting device including a plurality of cell blocks, each of which includes a plurality of light-emitting cells; and a plurality of controllers mounted on the printed circuit board, each of which is configured to drive a cell block corresponding thereto, from among the plurality of cell blocks. The plurality of cell blocks are electrically isolated from each other, the plurality of cell blocks include a first cell block and a second cell block, and a number of first light-emitting cells included in the first cell block is less than a number of second light-emitting cells included in the second cell block.

(7) According to an aspect of an example embodiment, a light-emitting device includes: a plurality of cell blocks electrically isolated from each other, each of which includes a plurality of light-emitting cells; and a plurality of pads configured to electrically connect the plurality of cell blocks to an external device. The plurality of cell blocks include a first cell block and a second cell block, and a number of first light-emitting cells included in the first cell block is less than a number of second light-emitting cells included in the second cell block.

(8) According to an aspect of an example embodiment, a light source module includes: a printed circuit board; a light-emitting device mounted on the printed circuit board, the light-emitting device including a plurality of cell blocks, each of which includes a plurality of light-emitting cells; and a plurality of controllers mounted on the printed circuit board, each of which is configured to drive a cell block corresponding thereto, from among the plurality of cell blocks. The plurality of cell blocks are electrically isolated from each other, and the plurality of cell blocks are arranged in a first row and a second row, and a number of light-emitting cells included in cell blocks arranged in

the first row is less than a number of light-emitting cells included in cell blocks arranged in the second row.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objects and features will become more apparent from the following description of example embodiments with reference to the accompanying drawings in which:

(2) FIG. **1** is a block diagram of a configuration of a light source module according to an example embodiment;

(3) FIG. **2** is a perspective view of a configuration of a light source module according to an example embodiment;

(4) FIG. **3** is a block diagram for explaining the connection of controllers to a light-emitting device in a light source module according to an example embodiment;

(5) FIG. **4** is a perspective view of a configuration of a light-emitting device according to an example embodiment;

(6) FIGS. **5** and **6** are circuit diagrams for explaining connection relationships between light-emitting cells in a light-emitting device according to an example embodiment;

(7) FIGS. **7** to **9** are diagrams of a light-emitting device according to example embodiments;

(8) FIGS. **10** and **11** are circuit diagrams for explaining a connection relationship between light-emitting cells included in a light-emitting device according to an example embodiment;

(9) FIGS. **12** and **13** are diagrams of light-emitting devices according to example embodiments;

(10) FIGS. **14** and **15** are cross-sectional views of light source modules, according to example embodiments;

(11) FIG. **16** is a schematic perspective view of a lighting apparatus including a light-emitting device, according to an example embodiment;

(12) FIG. **17** is a schematic perspective view of a flat-panel lighting apparatus including a light-emitting device, according to an example embodiment; and

(13) FIG. **18** is an exploded perspective view of a lighting apparatus including a light-emitting device, according to an example embodiment.

DETAILED DESCRIPTION

(14) The above and other aspects and features will become more apparent by describing example embodiments in detail with reference to the accompanying drawings. It will be understood that when an element or layer is referred to as being “over,” “above,” “on,” “connected to” or “coupled to” another element or layer, it can be directly over, above, on, connected or coupled to the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly over,” “directly above,” “directly on,” “directly connected to” or “directly coupled to” another element or layer, there are no intervening elements or layers present. The same reference numerals are used to denote the same elements in the drawings, and repeated descriptions thereof will be omitted.

(15) FIG. **1** is a block diagram of a configuration of a light source module **10** according to an example embodiment.

(16) Referring to FIG. **1**, the light source module **10** may include a light-emitting device **100** and a driving circuit **200**.

(17) The light-emitting device **100** may include a cell array CA including a plurality of light-emitting cells. The cell array CA included in the light-emitting device **100** may include a plurality of cell blocks BLK. In an example embodiment, the plurality of cell blocks BLK may be electrically isolated from each other.

(18) The driving circuit **200** may be connected to a power supply. The power supply may generate

an input voltage required for an operation of the light-emitting device **100**. In an example embodiment, when the light source module **10** is a headlamp for a vehicle, the power supply may include a battery mounted in the vehicle. In an example embodiment, when the light source module **10** is a household or industrial lighting fixture, the light source module **10** may further include an alternating-current (AC) power supply configured to generate an AC voltage, a rectifying circuit configured to rectify the AC voltage and generate a direct-current (DC) voltage, and a voltage regulating circuit.
(19) The driving circuit **200** may include a plurality of controllers **210**. Each of the plurality of controllers **210** may include an integrated circuit (IC).
(20) The plurality of controllers **210** may drive the cell array CA included in the light-emitting device **100**. In an example embodiment, each of the controllers **210** may be electrically connected to a corresponding one of the plurality of cell blocks BLK and control operations of light-emitting cells included in the corresponding cell block BLK. In an example embodiment, the number of controllers **210** may be equal to the number of cell blocks BLK included in the light-emitting device **100**. However, example embodiments are not limited thereto. For example, the number of controllers **210** may be different from the number of cell blocks BLK.
(21) FIG. **2** is a perspective view of a configuration of a light source module **10** according to an example embodiment.
(22) Referring to FIG. **2**, the light source module **10** may include a light-emitting device **100** and a plurality of controllers (e.g., first to ninth controllers **210_1** to **210_9**), which are mounted on a printed circuit board PCB. In an example embodiment, the light-emitting device **100** may include one chip, and the light source module **10** may include the light-emitting device **100** including one chip.
(23) The light-emitting device **100** may include a cell array CA in which a plurality of light-emitting cells are arranged in a matrix form. The cell array CA may include a plurality of cell blocks (e.g., first to ninth cell blocks BLK**1** to BLK**9**). Although nine cell blocks BLK**1** to BLK**9** are illustrated in FIG. **2**, example embodiments are not limited thereto, and the number and arrangement of cell blocks (e.g., BLK**1** to BLK**9**) may be changed.
(24) The first to ninth cell blocks BLK**1** to BLK**9** may be electrically isolated from each other. That is, light-emitting cells included in different cell blocks may be electrically isolated from each other.
(25) In an example embodiment, the light-emitting device **100** may have a rectangular shape having shorter sides in a second direction Y than in a first direction X. The first direction X and the second direction Y may be parallel to a main surface of a printed circuit board PCB and perpendicular to each other.
(26) In an example embodiment, in the light-emitting device **100**, the first to ninth cell blocks BLK**1** to BLK**9** may be arranged in a first row and a second row. In this case, each of the first row and the second row may refer to cell blocks arranged in parallel in the first direction X. Because the first to ninth cell blocks BLK**1** to BLK**9** are arranged in two rows, it may be structurally easy to form a plurality of pads configured to connect the light-emitting device **100** to the first to ninth controllers **210_1** to **210_9**.
(27) The light-emitting cells may be arranged in consideration of a light distribution type required for the light source module **10**. The number of light-emitting cells included in cell blocks arranged in the first row (e.g., BLK**1** to BLK**4**) may be less than the number of light-emitting cells included in cell blocks arranged in the second row (e.g., BLK**5** to BLK**9**). The number of cell blocks arranged in the first row may be different from the number of cell blocks arranged in the second row. For example, the first to fourth cell blocks BLK**1** to BLK**4** may be sequentially arranged in the first row in the first direction X, and the fifth to ninth cell blocks BLK**5** to BLK**9** may be sequentially arranged in the second row in an opposite direction to the first direction X.
(28) In an example embodiment, the number of light-emitting cells included in the first cell block BLK**1** arranged at one end of the first row may be less than the number of light-emitting cells

included in another one of the first to ninth cell blocks BLK**1** to BLK**9**. For example, the number of light-emitting cells in the first cell block BLK**1** may be less than the number of light-emitting cells in the second cell block BLK**2**, which is adjacent to the first cell block BLK**1**.

(29) In an example embodiment, the number of light-emitting cells included in the seventh cell block BLK**7** in the center of the second row may be less than the number of light-emitting cells included in another one of the first to ninth cell blocks BLK**1** to BLK**9**. For example, the number of light-emitting cells included in the seventh cell block BLK**7** may be less than the number of light-emitting cells included in the sixth cell block BLK**6**, which is adjacent to the seventh cell block BLK**7**, or be less than the number of light-emitting cells included in the eighth cell block BLK**8**, which is adjacent to the seventh cell block BLK**7**.

(30) In the light source module **10**, the light-emitting cells may be arranged in consideration of a light distribution type required for the light source module **10**. For example, when the light source module **10** is used as a headlamp for a vehicle, it may be relatively unnecessary to irradiate light to an upper outer area in front of a user in a direction in which the user travels. The first cell block BLK**1** and the fourth cell block BLK**4**, which are at both ends of the first row, may include a smaller number of light-emitting cells than other cell blocks, and thus, the light source module **10** may not separately irradiate light to unnecessary areas.

(31) In addition, for example, when the light source module **10** is used as the headlamp for the vehicle, the light source module **10** may need to irradiate light having a relatively high light intensity to a central area of a road in front of the user in the direction in which the user travels. The seventh cell block BLK**7** in the center of the second row may include a relatively small number of light-emitting cells. Even when the light source module **10** is controlled to irradiate light having a high intensity by applying a relatively large current to the seventh cell block BLK**7**, power consumption by the seventh cell block BLK**7** may be reduced because the seventh cell block BLK**7** includes fewer light-emitting cells.

(32) Each of the first to ninth controllers **210_1** to **210_9** may control light-emitting cells included in a cell block corresponding thereto, from among the first to ninth cell blocks BLK**1** to BLK**9**, to operate. For example, the first controller **210_1** may be electrically connected to the first cell block BLK**1** and control an operation of the first cell block BLK**1**. The second controller **210_2** may be electrically connected to the second cell block BLK**2** and control an operation of the second cell block BLK**2**. The description of the first and second controllers **210_1** and **210_2** may be equally applied to the third to ninth controllers **210_3** to **210_9**.

(33) In the light-emitting device **100**, the plurality of light-emitting cells in the cell array CA may be subdivided into the first to ninth cell blocks BLK**1** to BLK**9**, and operations of the first to ninth cell blocks BLK**1** to BLK**9** may be controlled by respectively different controllers. Thus, it may become easy to control a light emitting operation of the light-emitting device **100**. The control of brightness of the light-emitting device **100** may be subdivided, and a brightness adjusting speed may be increased.

(34) In an example embodiment, the first to ninth controllers **210_1** to **210_9** may be arranged in an order in which cell blocks corresponding to the first to ninth controllers **210_1** to **210_9** are arranged. For example, in the light-emitting device **100**, the first to fourth cell blocks BLK**1** to BLK**4** may be sequentially arranged in the first direction X, and the first to fourth controllers **210_1** to **210_4** corresponding to the first to fourth cell blocks BLK**1** to BLK**4** may also be sequentially arranged in the first direction X. Additionally, in the light-emitting device **100**, the fifth to ninth cell blocks BLK**5** to BLK**9** may be sequentially arranged in an opposite direction to the first direction X, and the fifth to ninth controllers **210_5** to **210_9** corresponding to the fifth to ninth cell blocks BLK**5** to BLK**9** may also be sequentially arranged in the opposite direction to the first direction X. In the light source module **10**, the first to ninth controllers **210_1** to **210_9** and the first to ninth cell blocks BLK**1** to BLK**9** may be arranged such that the order in which the first to ninth controllers **210_1** to **210_9** are arranged corresponds to the order in which the first to ninth cell blocks BLK**1**

to BLK**9**. Thus, it may become easy to form wirings or wires configured to electrically connect the first to ninth controllers **210_1** to **210_9** to the first to ninth cell blocks BLK**1** to BLK**9**.
(35) However, the arrangement of the plurality of controllers **210_1** to **210_9** is not limited thereto, and only some of the first to fourth controller **210_1-210_4** may be sequentially arranged in the first direction X, and only some of the fifth to ninth controllers **210_5** to **210_9** may be sequentially arranged in an opposite direction to the first direction X.
(36) In an example embodiment, the number of controllers (e.g., **210_1** to **210_9**) may be equal to the number of cell blocks (e.g., BLK**1** to BLK**9**). However, example embodiments are not limited thereto. At least two different controllers may be connected to one cell block and control the one cell block. Alternatively, one controller may control at least two different cell blocks.
(37) The light-emitting device **100** may be mounted on a central area CA_P of the printed circuit board PCB, and the first to ninth controllers **210_1** to **210_9** may be in first and second peripheral areas PA_P**1** and PA_P**2** of the printed circuit board PCB adjacent the light-emitting device **100**. For example, the first to fourth controllers **210_1** to **210_4** may be in the first peripheral area PA_P**1**, and the fifth to ninth controllers **210_5** to **210_9** may be in the second peripheral area PA_P**2**. Because the light-emitting device **100** is interposed between the first to ninth controllers **210_1** to **210_9** in the first and second peripheral areas PA_P**1** and PA_P**2** of the printed circuit board PCB, it may become easy to form wirings or wires configured to electrically connect the first to ninth controllers **210_1** to **210_9** to the first to ninth cell blocks BLK**1** to BLK**9**.
(38) In an example embodiment, the light-emitting device **100** may overlap the first to ninth controllers **210_1** to **210_9** in a direction (e.g., the first direction X or the second direction Y) parallel to the main surface of the printed circuit board PCB. In an example embodiment, the second peripheral area PA_P**2**, the central area CA_P, and the first peripheral area PA_P**1** may be sequentially arranged in the second direction Y.
(39) In the light source module **10**, one light-emitting device **100** including light-emitting cells configured to emit light may be in the central area CA_P of the printed circuit board PCB, and the first to ninth controllers **210_1** to **210_9**, which are configured to drive the light-emitting device **100** and respectively include separate chips, may be in the first and second peripheral areas PA_P**1** and PA_P**2**. Because the light-emitting device **100** including the plurality of light-emitting cells and the first to ninth controllers **210_1** to **210_9** are implemented as separate chips, designing the first to ninth controllers **210_1** to **210_9** may not be affected by the structures of the plurality of light-emitting cells. Accordingly, the design efficiency of the first to ninth controllers **210_1** to **210_9** may be increased.
(40) In addition, the light-emitting device **100** may be implemented as one LED chip and arranged in the central area CA_P of the printed circuit board PCB, and thus, light emitted from the light source module **10** may be concentrated in the central area CA_P. Because the emitted light is concentrated in the central area CA_P, the number of additional components (e.g., lenses) configured to condense the emitted light may be reduced. For instance, the light source module **10** may not include lenses. As the number of lenses included in the light source module **10** increases, the amount of light lost by the lenses may increase. Thus, the luminous efficiency of the light source module **10** may increase.
(41) The light source module **10** may further include an input unit configured to receive signals required for operations of the light source module **10** from the outside. The first to ninth controllers **210_1** to **210_9** may receive control signals from the input unit, and the first to ninth controllers **210_1** to **210_9** may control operations thereof in response to the control signals.
(42) In an example embodiment, the first to ninth controllers **210_1** to **210_9** may be electrically connected to each other in sequential order. For example, the first controller **210_1** may be electrically connected to the second controller **210_2**, the second controller **210_2** may be electrically connected to the first controller **210_1** and the third controller **210_3**, and the third controller **210_3** may be electrically connected to the second controller **210_2** and the fourth

controller **210_4**. The first controller **210_1** may receive a control signal from the input unit and transmit the control signal to the second controller **210_2**, and the second controller **210_2** may receive a control signal from the first controller **210_1** and transmit the control signal to the third controller **210_3**. The description of the first and second controllers **210_1** and **210_2** may be equally applied to the third to ninth controllers **210_3** to **210_9**.
(43) The printed circuit board PCB may include a metal and a metal compound. The printed circuit board PCB may be a metal-core printed circuit board (MCPCB), which includes, for example, copper (Cu).
(44) In an example embodiment, the printed circuit board PCB may be a flexible printed circuit board (FPCB), which may be flexible and easily modified into various shapes. In addition, the printed circuit board PCB may be a FR4-type PCB and include a resin material including epoxy, triazine, silicon, and polyimide or a ceramic material, such as silicon nitride, aluminum nitride (AlN) and aluminum oxide (Al.sub.2O.sub.3).
(45) FIG. **3** is a block diagram for explaining the connection of controllers to a light-emitting device **100** in a light source module **10** according to an example embodiment. Only a first cell block and a second cell block connected to a first controller and a second controller are illustrated in FIG. **3** for brevity, and the same description may be applied to other controllers (e.g., the third to ninth **210_3** to **210_9**) and other cell blocks (e.g., the third to ninth BLK**3** to BLK**9**).
(46) Referring to FIG. **3**, the light-emitting device **100** may include a first cell block BLK**1** including a plurality of first light-emitting cells and a second cell block BLK**2** including a plurality of second light-emitting cells. The light-emitting device **100** may include a plurality of first pads PAD**1** connected to the first cell block BLK**1** and a plurality of second pads PAD**2** connected to the second cell block BLK**2**.
(47) In an example embodiment, the number of first light-emitting cells in the first cell block BLK**1** may be less than the number of second light-emitting cells in the second cell block BLK**2**. The number of first pads PAD**1** connected to the first cell block BLK**1** may be less than the number of second pads PAD**2** connected to the second cell block BLK**2**. Although FIG. **3** illustrates a case in which the number of first pads PAD**1** is one less than the number of second pads PAD**2**, example embodiments are not limited thereto, and the number of first pads PAD**1** may at least two less than the number of second pads PAD**2**.
(48) A first controller **210_1** may be connected to a plurality of first channels CH**11** to CH**1***n* and drive the first cell block BLK**1**, which is a cell block corresponding to the first controller **210_1**, through at least some (e.g., CH**11** to CH**1***n*-**1**) of the plurality of first channels CH**11** to CH**1***n*. The first controller **210_1** may apply a voltage to the first cell block BLK**1** corresponding thereto through the at least some first channels CH**11** to CH**1***n*-**1** and adjust the intensity of light emitted from the first cell block BLK**1**.
(49) A second controller **210_2** may be connected to a plurality of second channels CH**21** to CH**2***n* and drive a second cell block BLK**2**, which is a cell block corresponding to the second controller **210_2**, through at least some of the plurality of second channels CH**21** to CH**2***n*. The second controller **210_2** may apply a voltage to the second cell block BLK**2** corresponding thereto through the at least some of the second channels CH**21** to CH**2***n* and adjust the intensity of light emitted from the second cell block BLK**2**.
(50) In an example embodiment, the plurality of first channels CH**11** to CH**1***n* connected to the first controller **210_1** may be different channels from the plurality of second channels CH**21** to CH**2***n* connected to the second controller **210_2**. Accordingly, the first controller **210_1** and the second controller **210_2**, which are different controllers, may individually control cell blocks corresponding respectively thereto.
(51) In an example embodiment, the first controller **210_1** and the second controller **210_2** may be connected to the same number of channels. For example, the first controller **210_1** may be connected to n first channels CH**11** to CH**1***n* and apply a voltage to each of the first channels CH**11**

to CH**1***n*. The second controller **210_2** may be connected to n second channels CH**21** to CH**2***n* and apply a voltage to each of the second channels CH**21** to CH**2***n*.
(52) In an example embodiment, the number (e.g., (n−1)) of first pads PAD**1** may be less than the number (e.g., n) of first channels CH**11** to CH**1***n* connected to the first controller **210_1**. At least one channel (e.g., CH**1***n*) of the plurality of first channels CH**11** to CH**1***n* may not be connected to the plurality of first pads PAD**1** but be electrically isolated from the first cell block BLK**1**. Accordingly, the number of channels connected to the first to ninth controllers **210_1** to **210_9** of the light source module **10** of FIG. **2** may not correspond to the number of pads formed in the light-emitting device **100**.
(53) However, example embodiments are not limited to the illustration of FIG. **3**. Because the number of first light-emitting cells included in the first cell block BLK**1** is less than the number of second light-emitting cells included in the second cell block BLK**2**, the number of first channels CH**11** to CH**1***n* connected to the first controller **210_1** configured to control an operation of the first cell block BLK**1** may be less than the number of second channels CH**21** to CH**2***n* connected to the second controller **210_2** configured to control an operation of the second cell block BLK**2**.
(54) FIG. **4** is a perspective view of a configuration of a light-emitting device **100** according to an example embodiment.
(55) Referring to FIG. **4**, the light-emitting device **100** may include a cell array region PXR in which a cell array CA is formed and first and second pad regions PDR**1** and PDR**2** in which a plurality of pads PAD are formed. The cell array region PXR may be in a central area of the light-emitting device **100**, and the first and second pad regions PDR**1** and PDR**2** may be in a peripheral area surrounding the central area. For instance, the second pad region PDR**2**, the cell array region PXR, and the first pad region PDR**1** may be sequentially arranged in a second direction Y.
(56) The cell array CA may include a plurality of cell blocks (e.g., first to ninth cell blocks BLK**1** to BLK**9**). In this case, the first to ninth cell blocks BLK**1** to BLK**9** may be electrically isolated from each other. Although a total of nine cell blocks BLK**1** to BLK**9** are illustrated in FIG. **4**, the light-emitting device **100** is not limited thereto, and the number and arrangement of cell blocks (e.g., BLK**1** to BLK**9**) may be changed.
(57) The first to ninth cell blocks BLK**1** to BLK**9** may be arranged in a first row and a second row, which are arranged in parallel to each other in the second direction Y. For example, the first to fourth cell blocks BLK**1** to BLK**4** may be arranged in the first row, and the fifth to ninth cell blocks BLK**5** to BLK**9** may be arranged in a second row. At least one of the plurality of cell blocks BLK**1** to BLK**9** may include a different number of light-emitting cells **111** from other cell blocks. In an example embodiment, the first cell block BLK**1** and the fourth cell block BLK**4** at both ends of the first row may include a smaller number light-emitting cells **111** than at least one of other cell blocks. For example, the first cell block BLK**1** and the fourth cell block BLK**4**, which are at both ends of the cell array CA, may include a smaller number of light-emitting cells **111** than other cell blocks positioned adjacent thereto. The first cell block BLK**1** may include a smaller number of light-emitting cells **111** than the second cell block BLK**2**, which is adjacent to the first cell block BLK**1** in a first direction X. Alternatively, the first cell block BLK**1** may include a smaller number of light-emitting cells **111** than the ninth cell block BLK**9**, which is adjacent to the first cell block BLK**1** in an opposite direction to the second direction Y. The fourth cell block BLK**4** may include a smaller number of light-emitting cells **111** than the third cell block BLK**3**, which is adjacent to the fourth cell block BLK**4** in an opposite direction to the first direction X. Alternatively, the fourth cell block BLK**4** may include a smaller number of light-emitting cells **111** than the fifth cell block BLK**5**, which is adjacent to the fourth cell block BLK**4** in an opposite direction to the second direction Y.
(58) In an example embodiment, the seventh cell block BLK**7** in the center of the cell array CA may include a smaller number of light-emitting cells **111** than at least one of other cell blocks. For example, the seventh cell block BLK**7** may include a smallest number of light-emitting cells **111**

from among the first to ninth cell blocks BLK**1** to BLK**9**.
(59) FIG. **4** illustrates an example in which each of the second, third, fifth, sixth, eighth, and ninth cell blocks BLK**2**, BLK**3**, BLK**5**, BLK**6**, BLK**8**, and BLK**9** includes twelve light-emitting cells **111**, each of the first and fourth cell blocks BLK**1** and BLK**4** includes eleven light-emitting cells, and the seventh cell block BLK**7** includes eight light-emitting cells, however example embodiments are not limited thereto. In this regard, number of light-emitting cells **111** included in each of the first to ninth cell blocks BLK**1** to BLK**9** may be variously selected.
(60) In an example embodiment, from among light-emitting cells included in cell blocks arranged in the first row, light-emitting cells adjacently arranged in the second direction Y may be driven (or turned on) or may not be driven (or turned off) simultaneously. In an example embodiment, partition walls may not be formed between the light-emitting cells adjacently arranged in the second direction Y, from among the light-emitting cells included in the cell blocks arranged in the first row. In an example embodiment, the first to ninth cell blocks BLK**1** to BLK**9** may be arranged in a rectangular shape such that the length L**1** of the light-emitting device **100** in the first direction X is greater than a length L**2** of the light-emitting device **100** in the second direction Y. For example, the first to ninth cell blocks BLK**1** to BLK**9** may be arranged in a total of two rows, for example, a first row and a second row.
(61) In an example embodiment, the length L**1** of the light-emitting device **100** in the first direction X may be about 1.1 times or more the length L**2** of the light-emitting device **100** in the second direction Y. In an example embodiment, the length L**1** of the light-emitting device **100** in the first direction X may be about 100 times or less the length L**2** of the light-emitting device **100** in the second direction Y. According to an example embodiment, a thickness of the light-emitting device **100** (i.e., a length of the light-emitting device **100** in a third direction Z) may be several tens of μm to several hundreds of μm and may be less than or equal to about 1/10 of the length L**1** of the light-emitting device **100** in the first direction X. Because the light-emitting device **100** having the above-described dimensions has dimensions optimized for resistance to physical stress, the warpage of the light-emitting device **100** may be minimized.
(62) The first to fourth cell blocks BLK**1** to BLK**4** may be sequentially arranged in the first row in the first direction X, and the fifth to ninth cell blocks BLK**5** to BLK**9** may be sequentially arranged in the second row in an opposite direction to the first direction X. Although FIG. **4** illustrates the first to ninth cell blocks BLK**1** to BLK**9** arranged in two rows, example embodiments are not limited thereto, and the cell array CA may include the first to ninth cell blocks BLK**1** to BLK**9** arranged in at least three rows.
(63) The first to fourth cell blocks BLK**1** to BLK**4** arranged in the first row may be electrically connected to first to fourth controllers (e.g., **210_1** to **210_4** of FIG. **2**) through the plurality of pads PAD arranged in the first pad region PDR**1**. The fifth to ninth cell blocks BLK**5** to BLK**9** arranged in the second row may be electrically connected to fifth to ninth controllers (e.g., **210_5** to **210_9** of FIG. **2**) through the plurality of pads PAD arranged in the second pad region PDR**2**.
(64) In the light-emitting device **100**, the first and second pad regions PDR**1** and PDR**2** may not be in the cell array region PXR but be arranged in parallel with each other and extend in the second direction Y. That is, the first and second pad regions PDR**1** and PDR**2** in which the plurality of pads PAD are arranged may not overlap the cell array region PXR in the third direction Z, which is perpendicular to a main surface of a substrate. Because the light-emitting device **100** includes the first and second pad regions PDR**1** and PDR**2** separately from the cell array region PXR, the density of the light-emitting cells **111** in the cell array region PXR may be increased. Furthermore, a plurality of pads PAD may be in the peripheral area of the light-emitting device **100**, and thus, it may be easy to form components (e.g., bonding wires) configured to driving chips to the plurality of pads PAD.
(65) In example embodiments, in a view from above, an area of the cell array region PXR may be in a range of about 50% to about 90% of a total area of the light-emitting device **100**, and an area of

the first and second pad regions PDR**1** and PDR**2** may be in a range of about 10% to about 50% of the total area of the light-emitting device **100**, without being limited thereto.
(66) FIGS. **5** and **6** are circuit diagrams for explaining connection relationships between light-emitting cells in a light-emitting device **100** according to an example embodiment. FIGS. **5** and **6** are equivalent circuit diagrams of the first cell block BLK**1** and the second cell block BLK**2** of FIG. **4**. In FIGS. **5** and **6**, one light-emitting cell may correspond to one diode. The description of the first and second cell blocks BLK**1** and BLK**2** of FIGS. **5** and **6** may be equally applied to the third to ninth cell blocks BLK**3** to BLK**9** of FIG. **4**.
(67) Referring to FIGS. **4** and **5**, the first cell block BLK**1** may include a plurality of first light-emitting cells **111_1**, each of which is implemented as an LED, and the second cell block BLK**2** may include a plurality of second light-emitting cells **111_2**, each of which is implemented as an LED. The number of first light-emitting cells **111_1** may be less than the number of second light-emitting cells **111_2**. The first cell block BLK**1** and the second cell block BLK**2** may be electrically insulated from each other, and operations of the first cell block BLK**1** and the second cell block BLK**2** may be controlled by different controllers.
(68) Cathodes or anodes of the light-emitting cells (refer to **111** in FIG. **4**) included in each of the first to ninth cell blocks BLK**1** to BLK**9** may be electrically connected to each other. For example, cathodes or anodes of the plurality of first light-emitting cells **111_1** included in the first cell block BLK**1** may be electrically connected to each other.
(69) In an example embodiment, the light-emitting cells **111** included in each of the first to ninth cell blocks BLK**1** to BLK**9** may be connected in series to other light-emitting cells **111** within the same block. For example, the plurality of first light-emitting cells **111_1** in the first cell block BLK**1** may be connected in series to each other, and the plurality of second light-emitting cells **111_2** in the second cell block BLK**2** may be connected in series to each other. The plurality of first light-emitting cells **111_1** in the first cell block BLK**1** may be electrically isolated from the plurality of second light-emitting cells **111_2** in the second cell block BLK**2**. Both ends of each of the plurality of first light-emitting cells **111_1** and the plurality of second light-emitting cells **111_2** may be respectively connected to different pads.
(70) Because the plurality of first light-emitting cells **111_1** are connected in series to each other and the plurality of second light-emitting cells **111_2** are connected in series to each other, when voltages applied to nodes at which the plurality of first light-emitting cells **111_1** are connected to each other and nodes at which the plurality of second light-emitting cells **111_2** are connected to each other are controlled, operations of the plurality of first light-emitting cells **111_1** and the plurality of second light-emitting cells **111_2** may be controlled. Accordingly, the number of first pads PAD**1** connected to the plurality of first light-emitting cells **111_1** may be less than twice the number of first light-emitting cells **111_1**, and the number of second pads PAD**2** connected to the plurality of second light-emitting cells **111_2** may be less than twice the number of second light-emitting cells **111_2**.
(71) A first controller **210_1** may be electrically connected to a plurality of first pads PAD**1**. The first controller **210_1** may adjust voltages applied to two different pads, from among the plurality of first pads PAD**1**, and thus, one light-emitting cell of which a cathode and an anode are respectively connected to the two pads may be driven. For example, the first controller **210_1** may adjust a brightness of each of the first light-emitting cells **111_1** using a pulse width modulation (PWM) scheme. That is, the first controller **210_1** may adjust the brightness of each of the plurality of first light-emitting cells **111_1** by modulating a pulse width of a voltage applied to each of the plurality of first pads PAD**1**.
(72) A second controller **210_2** may be electrically connected to a plurality of second pads PAD**2**. The second controller **210_2** may adjust voltages applied to two different pads, from among the plurality of second pads PAD**2**, and thus, one light-emitting cell of which a cathode and an anode are respectively connected to the two pads may be driven.

(73) According to an order in which the first cell block BLK**1** and the second cell block BLK**2** are arranged, the plurality of first pads PAD**1** and the plurality of second pads PAD**2** corresponding respectively to the first cell block BLK**1** and the second cell block BLK**2** may be arranged in sequential order. The first cell block BLK**1** and the plurality of first pads PAD**1** may be arranged in parallel and extend along a second direction Y, and the second cell block BLK**2** and the plurality of second pads PAD**2** may be arranged in parallel and extend along the second direction Y.
(74) Referring to FIG. **6**, a first cell block BLK**1**′ may further include a plurality of first light-emitting cells **111_1** and a dummy light-emitting cell **111_1**′, each of which is implemented as an LED. The dummy light-emitting cell **111_1**′ may not be connected to the plurality of first pads PAD**1** and may not substantially operate. In an example embodiment, the number of first light-emitting cells **111_1** and the dummy light-emitting cell **111_1**′ included in the first cell block BLK**1**′ may be equal to the number of second light-emitting cells **111_2** included in the second cell block BLK**2**.
(75) FIGS. **7** to **9** are diagrams of a light-emitting device **100** according to example embodiments. FIG. **7** is an enlarged plan view of region BX of FIG. **4**. FIG. **7** illustrates an example of the light-emitting device **100**, in which a first cell block does not include a dummy light-emitting cell as shown in FIG. **5**. FIG. **8** is an enlarged cross-sectional view of configurations of some components, which is taken along line II-II′ of FIG. **7**, according to an example embodiment. FIG. **9** is an enlarged view of region CX**2** of FIG. **8**.
(76) Referring to FIGS. **7** to **9**, a partition wall structure WS may be on a plurality of light-emitting structures **120**. As shown in FIG. **7**, the partition wall structure WS may include a plurality of partition walls WSI and an outer partition wall WSO. The plurality of partition walls WSI may define a plurality of cell spaces PXS in a cell array region PXR, and the outer partition wall WSO may be on outermost edges of the plurality of partition walls WSI. Light-emitting cells may be respectively disposed in the plurality of cell spaces PXS.
(77) The partition wall structure WS may include round-corner sidewall units PWC facing the plurality of cell spaces PXS. The plurality of cell spaces PXS having corners rounded by the round-corner sidewall units PWC of the partition wall structure WS may be respectively defined.
(78) Each of the plurality of partition walls WSI may have a first width w**11** of about 10 μm to about 100 μm in a lateral direction (i.e., a second direction Y). The outer partition wall WSO may have a second width w**12** of about 10 μm to about 1 mm in the lateral direction (i.e., the second direction Y). The partition wall structure WS may be formed such that the outer partition wall WSO is formed to have the second width w**12** greater than the first width w**11** of the plurality of partition walls WSI. Thus, the structural stability of the light-emitting device **100** may be improved. For example, even when repetitive vibration and impact are applied to the light-emitting device **100** when the light-emitting device **100** is used as a headlamp for a vehicle, the reliability of the light-emitting device **100** may be improved by excellent structural stability between the partition wall structures WS and the fluorescent layer **160** positioned within the partition wall structures WS.
(79) Each of the plurality of light-emitting structures **120** may include a first conductive semiconductor layer **122**, an active layer **124**, and a second conductive semiconductor layer **126**. An insulating liner **132**, a first contact **134**A, a second contact **134**B, and a wiring structure **140** may be on a bottom surface of each of the plurality of light-emitting structures **120**.
(80) For brevity, as shown in FIG. **9**, a surface of the light-emitting structure **120**, which faces the plurality of partition walls WSI, may be referred to as a top surface of the light-emitting structure **120**, while a surface of the light-emitting structure **120** opposite to the top surface of the light-emitting structure **120** (i.e., a surface of the light-emitting structure **120** positioned far from the plurality of partition walls WSI) may be referred to as a bottom surface of the light-emitting structure **120**. For example, the first conductive semiconductor layer **122**, the active layer **124**, and the second conductive semiconductor layer **126** may be stacked in the vertical direction (i.e., a third direction (Z)) from the top surface of the light-emitting structure **120** to the bottom surface thereof.

Thus, the top surface of the light-emitting structure **120** may correspond to a top surface of the first conductive semiconductor layer **122**, and the bottom surface of the light-emitting structure **120** may correspond to a bottom surface of the second conductive semiconductor layer **126**.
(81) The first conductive semiconductor layer **122** may be a nitride semiconductor having a composition of n-type In.sub.xAl.sub.yGa.sub.(1-x-y)N (where 0≤x<1, 0≤y<1, and 0≤x+y<1). For example, the n-type impurities may be silicon (Si). For example, the first conductive semiconductor layer **122** may include GaN containing n-type impurities.
(82) In example embodiments, the first conductive semiconductor layer **122** may include a first conductive semiconductor contact layer and a current diffusion layer. An impurity concentration of the first conductive semiconductor contact layer may be in a range of 2×10.sup.18 cm.sup.−3 to 9×10.sup.19 cm.sup.−3. A thickness of the first conductive semiconductor contact layer may be about 1 μm to about 5 μm. The current diffusion layer may have a structure in which a plurality of In.sub.xAl.sub.yGa.sub.(1-x-y)N layers (where 0≤x, y<1, and 0≤x+y<1) having different compositions or different impurity contents are alternately stacked. For example, the current diffusion layer may have an n-type superlattice structure in which n-type GaN layers and/or Al.sub.xIn.sub.yGa.sub.zN layers (where 0≤x,y,z≤1, and x+y+z≠0) each having a thickness of about 1 nm to about 500 nm are alternately stacked. An impurity concentration of the current diffusion layer may be in the range of 2×10.sup.18 cm.sup.−3 to 9×10.sup.19 cm.sup.−3.
(83) The active layer **124** may be interposed between the first conductive semiconductor layer **122** and the second conductive semiconductor layer **126**. The active layer **124** may discharge light having some energy by recombination of electrons and holes during the driving of the light-emitting device **100**. The active layer **124** may have a multiple quantum well (MQW) structure in which quantum well layers and quantum barrier layers are alternately stacked. For example, each of the quantum well layers and each of the quantum barrier layers may include In.sub.xAl.sub.yGa.sub.(1-x-y)N layers (where 0≤x, y≤1, and 0≤x+y≤1) having different compositions. For example, the quantum well layer may include In.sub.xGa.sub.1-xN (where 0≤x≤1), and the quantum barrier layer may include GaN or AlGaN. Thicknesses of the quantum well layer and the quantum barrier layer may be in the range of about 1 nm to about 50 nm. The active layer **124** is not limited to having the MQW structure and may have a single quantum well structure.
(84) The second conductive semiconductor layer **126** may include a nitride semiconductor layer having a composition of p-type In.sub.xAl.sub.yGa.sub.(1-x-y)N (where 0≤x<1, 0≤y<1, and 0≤x+y≤1). For example, the p-type impurities may be magnesium (Mg).
(85) In example embodiments, the second conductive semiconductor layer **126** may include an electron blocking layer, a low-concentration p-type GaN layer, and a high-concentration p-type GaN layer, which are stacked in a vertical direction. For example, the electron blocking layer may have a structure in which a plurality of In.sub.xAl.sub.yGa.sub.(1-x-y)N layers (where 0≤x, y≤1, and 0≤x+y≤1) having a thickness of about 5 nm to about 100 nm and having different compositions are alternately stacked, or may include a single layer including Al.sub.yGa.sub.(1-y)N (where 0<y≤1). An energy band gap of the electron blocking layer may be reduced in a direction away from the active layer **124**. For example, aluminum (Al) content in the electron blocking layer may be reduced in the direction away from the active layer **124**.
(86) Each of the plurality of light-emitting structures **120** may be spaced apart from light-emitting structures **120** adjacent thereto with a device isolation region IA interposed therebetween. A distance s**11** between the plurality of light-emitting structures **120** may be less than the first width w**11** of each of the plurality of partition walls WSI, without being limited thereto.
(87) The insulating liner **132** may be positioned to conformally cover an inner wall of the device isolation region IA and a side surface of each of the plurality of light-emitting structures **120**. Also, the insulating liner **132** may be on an inner wall of the opening E, which completely passes through the active layer **124** and the second conductive semiconductor layer **126**. In example embodiments,

the insulating liner **132** may include silicon oxide, silicon oxynitride, or silicon nitride. In some example embodiments, the insulating liner **132** may have a structure in which a plurality of insulating layers are stacked.
(88) The first contact **134**A may be connected to the first conductive semiconductor layer **122** in the opening E penetrating the active layer **124** and the second conductive semiconductor layer **126**. The second contact **134**B may be on the bottom surface of the second conductive semiconductor layer **126**. The insulating liner **132** may electrically insulate the first contact **134**A from the active layer **124** and the second conductive semiconductor layer **126**. The insulating liner **132** may be interposed between the first contact **134**A and the second contact **134**B on the bottom surface of the second conductive semiconductor layer **126** and electrically insulate the first contact **134**A from the second contact **134**B. Each of the first contact **134**A and the second contact **234**B may include silver (Ag), aluminum (Al), nickel (Ni), chromium (Cr), gold (Au), platinum (Pt), palladium (Pd), tin (Sn), tungsten (W), rhodium (Rh), iridium (Ir), ruthenium (Ru), magnesium (Mg), zinc (Zn), titanium (Ti), copper (Cu), or a combination thereof. Each of the first contact **134**A and the second contact **134**B may include a metal material having high reflectivity.
(89) A lower reflective layer **136** may be on the insulating liner **132** positioned on the inner wall of the device isolation region IA. The lower reflective layer **136** may reflect light emitted from sidewalls of the plurality of light-emitting structures **120** and direct the reflected light into the plurality of cell spaces PXS.
(90) In example embodiments, the lower reflective layer **136** may include Ag, Al, Ni, Cr, Au, Pt, Pd, Sn, W, Rh, Ir, Ru, Mg, Zn, Ti, Cu, or a combination thereof. The lower reflective layer **136** may include a metal material having high reflectivity. In other example embodiments, the lower reflective layer **136** may include a distributed Bragg reflector (DBR). For example, the DBR may have a structure in which a plurality of insulating layers having different refractive indexes are repeatedly stacked. Each of the insulating layers in the DBR may include oxide, nitride, or a combination thereof, for example, SiO.sub.2, SiN, SiO.sub.xN.sub.y, TiO.sub.2, Si.sub.3N.sub.4, Al.sub.2O.sub.3, TiN, AlN, ZrO.sub.2, TiAlN, or TiSiN.
(91) A wiring structure **140** may be on the insulating liner **132**, the first contact **134**A, the second contact **134**B, and the lower reflective layer **136**. The wiring structure **140** may include a plurality of insulating layers **142** and a plurality of wiring layers **144**. The plurality of wiring layers **144** may electrically connect each of the first contact **134**A and the second contact **134**B to the pad PAD. Some of the plurality of wiring layers **144** may be on the inner wall of the device isolation region IA, and the plurality of insulating layers **142** may respectively cover the plurality of wiring layers **144** and fill the device isolation region IA. As shown in FIG. **9**, the plurality of wiring layers **144** may include at least two wiring layers **144** located at different levels in the vertical direction, but example embodiments are not limited thereto. Each of the plurality of wiring layers **144** may include Ag, Al, Ni, Cr, Au, Pt, Pd, Sn, W, Rh, Ir, Ru, Mg, Zn, Ti, Cu, or a combination thereof.
(92) In example embodiments, the plurality of wiring layers **144** may electrically connect the first contact **134**A of one light-emitting structure **120** to the second contact **134**B of another light-emitting structure **120** connected in series to the one light-emitting structure **120** so that the plurality of light-emitting cells included in one cell block may be connected in series to each other.
(93) Alternatively, in contrast to the illustration of FIG. **8**, in another example embodiment, the plurality of wiring layers **144** may electrically connect the first contact **134**A of one light-emitting structure **120** to the first contact **134**A of another light-emitting structure **120** connected in parallel to the one light-emitting structure **120** so that the plurality of light-emitting cells included in one cell block may be connected in parallel to each other.
(94) The pad PAD may be in the pad region PDR and connected to the plurality of wiring layers **144**. The pad PAD may be at a lower level than the partition wall structure WS. In example embodiments, a sidewall and a bottom surface of the pad PAD may be covered by the plurality of insulating layers **142**, and a top surface of the pad PAD may be at a lower level than the top

surfaces of the plurality of light-emitting structures **120**. In other example embodiments, in contrast to the illustration of FIG. **8**, some of the plurality of light-emitting structures **120** may be in the pad region PDR, and the pad PAD may be in openings formed in the plurality of light-emitting structures **120**. In this case, the top surface of the pad PAD may be at the same level as the top surfaces of the plurality of light-emitting structures **120**. Connection members (e.g., bonding wires) for electrical connection with a controller may be on the pad PAD.
(95) The partition wall structures WS may be on the top surfaces of the plurality of light-emitting structures **120**. The partition wall structures WS may include silicon (Si), silicon carbide (SiC), sapphire (Al_2O_3), or gallium nitride (GaN). In an example process, after the plurality of light-emitting structures **120** are formed on a substrate (refer to **110** in FIG. **18**A), the partition wall structures WS may be formed by removing portions of the substrate **110**. In this case, the partition wall structures WS may be portions of the substrate **110**, which serve as a growth substrate for forming the light-emitting structures **120**.
(96) The plurality of partition walls WSI may be arranged in a matrix form in a plan view, and the plurality of cell spaces PXS may be defined by the plurality of partition walls WSI. Each of the plurality of partition walls WSI may include a recess region RS, which is at a bottom of each of the plurality of partition walls WSI to vertically overlap the device isolation region IA. The recess region RS may be formed by removing a portion of the substrate **110** during an etching process for separating a light-emitting stack into the plurality of light-emitting structures **120**. The insulating liner **132** may be arranged to contact the recess region RS on a bottom surface of each of the plurality of partition walls WSI.
(97) The top surfaces of the plurality of light-emitting structures **120** may be exposed at bottoms of the plurality of cell spaces PXS. For example, concave/convex portions **120**P may be formed in the top surfaces of the plurality of light-emitting structures **120** positioned at the bottoms of the plurality of cell spaces PXS. Light extraction efficiency of the plurality of light-emitting structures **120** may be improved by the concave/convex portions **120**P, but example embodiments are not limited thereto.
(98) A passivation structure **150** may be on a top surface WST and a sidewall WSS of each of the plurality of partition walls WSI. The passivation structure **150** may include a first passivation layer **152** and a second passivation layer **154**, which are conformally on the top surface WST and the sidewall WSS of each of the plurality of partition walls WSI. The passivation structure **150** may also be conformally on the top surfaces of the light-emitting structures **120** (e.g., on the concave/convex portions **120**P) positioned at the bottoms of the plurality of cell spaces PXS. Although FIG. **8** illustrates two passivation layers (e.g., **152** and **154**), example embodiments are not limited thereto, and the passivation structure **150** may include three or more passivation layers.
(99) In example embodiments, the first passivation layer **152** may include a first insulating material, and the second passivation layer **154** may include a second insulating material that is different from the first insulating material. Each of the first insulating material and the second insulating material may include at least one of silicon oxide, silicon nitride, silicon oxynitride, aluminum oxide, and aluminum nitride.
(100) In example embodiments, the passivation structure **150** may include a first portion **150**P**1** positioned on the top surface WST of each of the plurality of partition walls WSI, a second portion **150**P**2** positioned on the sidewall WSS of each of the plurality of partition walls WSI, and a third portion **150**P**3** positioned on the top surfaces of the plurality of light-emitting structures **120**. In some example embodiments, a first thickness t**11** of the first portion **150**P**1** may be less than or equal to a second thickness t**12** of the second portion **150**P**2**. Also, a third thickness t**13** of the third portion **150**P**3** may be less than or equal to the second thickness t**12** of the second portion **150**P**2**. In some example embodiments, the first thickness t**11** of the first portion **150**P**1** may be in a range of about 0.1 μm to about 2 μm, and the second thickness t**12** of the second portion **150**P**2** may be in a range of about 0.5 μm to about 5 μm.

(101) As shown in FIG. **9**, portions of the first passivation layer **152** included in the first portion **150**P**1** (i.e., portions of the first passivation layer **152** on the top surfaces WST of the plurality of partition wall structures WS) may have a thickness less than a thickness of portions of the first passivation layer **152** included in the second portion **150**P**2** (i.e., portions of the first passivation layer **152** on the sidewalls WSS of the plurality of partition wall structures WS). Similarly, portions of the second passivation layer **154** included in the first portion **150**P**1** (i.e., portions of the second passivation layer **154** on the top surfaces WST of the plurality of partition wall structures WS) may have a thickness less than a thickness of portions of the second passivation layer **154** included in the second portion **150**P**2** (i.e., portions of the second passivation layer **154** on the sidewalls WSS of the plurality of partition wall structures WS).
(102) In example embodiments, the first passivation layer **152** may have a relatively uniform thickness on the sidewall WSS of each of the plurality of partition walls WSI. Here, the expression "relatively uniform thickness" may mean that a minimum thickness of the first passivation layer **152** positioned on the sidewall WSS of each of the plurality of partition walls WSI is within about 10% of a maximum thickness thereof. Also, the second passivation layer **154** may have a relatively uniform thickness on the sidewall WSS of each of the plurality of partition walls WSI. In an example manufacturing process, the first and second passivation layers **152** and **154** may be formed using a material having excellent step coverage characteristics or a manufacturing process (e.g., an atomic layer deposition (ALD) process), which is advantageous in forming materials having excellent step coverage characteristics.
(103) For example, the first thickness t**11** may be less than a critical thickness for the passivation structure **150** to act as a light guide. For example, when the first thickness t**11** of the first portion **150**P**1** of the passivation structure **150** positioned on the top surface WST of each of the plurality of partition walls WSI is greater than the critical thickness, light emitted from one light-emitting cell may be directed into an adjacent light-emitting cell through the first portion **150**P**1** of the passivation structure **150**. Accordingly, when one light-emitting cell is turned on, light may be absorbed or penetrated into a light-emitting cell adjacent thereto, and thus, the adjacent light-emitting cell may be difficult to be put into a complete off state. When the first thickness t**11** of the first portion **150**P**1** is less than the critical thickness, light emitted from one light-emitting cell may be blocked from an adjacent light-emitting cell through the first portion **150**P**1** of the passivation structure **150**. The first thickness t**11** of the first portion **150**P**1** may be less than or equal to the second thickness t**12** of the second portion **150**P**2**. Particularly, the first thickness t**11** of the first portion **150**P**1** may be less than the critical thickness for the passivation structure **150** to act as the light guide. Accordingly, the second portion **150**P**2** of the passivation structure **150** may provide a sufficient thickness to prevent contamination of the fluorescent layer **160**, while undesired light crosstalk between adjacent pixels PX due to the first portion **150**P**1** of the passivation structure **150** may be prevented.
(104) A fluorescent layer **160** may be inside the plurality of cell spaces PXS on the top surfaces of the plurality of light-emitting structures **120**. As shown in FIG. **8**, the fluorescent layer **160** may substantially fill the total space of the plurality of cell spaces PXS on the passivation structure **150**. A top surface of the fluorescent layer **160** may be at the same level as the top surface WST of each of the plurality of partition walls WSI, but example embodiments are not limited thereto.
(105) The fluorescent layer **160** may include a single material capable of converting the color of light emitted from the light-emitting structure **120** into a desired color. That is, a fluorescent layer **160** associated with the same color may be in the plurality of cell spaces PXS. However, example embodiments are not limited thereto. The color of a fluorescent layer **160** in some of the plurality of cell spaces PXS may be different from the color of a fluorescent layer **160** in the remaining cell spaces PXS.
(106) The fluorescent layer **160** may include a resin containing a fluorescent material dispersed therein or a film containing a fluorescent material. For example, the fluorescent layer **160** may

include a fluorescent material film in which fluorescent material particles are uniformly dispersed at a certain concentration. The fluorescent material particles may be a wavelength conversion material that changes the wavelength of light emitted from the plurality of light-emitting structures **120**. The fluorescent layer **160** may include at least two kinds of fluorescent material particles having different size distributions to improve the density and color uniformity of the fluorescent material particles.

(107) In example embodiments, the fluorescent material may have various colors and various compositions such as an oxide-based composition, a silicate-based composition, a nitride-based composition, and a fluoride-based composition. For example, β-SiAlON:Eu.sup.2+ (green), (Ca, Sr)AlSiN.sub.3:Eu.sup.2+ (red), La.sub.3Si.sub.6N.sub.11:Ce.sup.3+ (yellow), K.sub.2SiF.sub.6:Mn.sub.4.sup.+(red), SrLiAl.sub.3N.sub.4:Eu(red), Ln.sub.4-x(Eu.sub.zM.sub.1-z).sub.xSi.sub.12-yAl.sub.yO.sub.3+x+yN.sub.18-x-y(0.5≤x≤3, 0<z<0.3, 0<y≤4)(red), K.sub.2TiF.sub.6:Mn.sub.4.sup.+(red), NaYF.sub.4:Mn.sub.4.sup.+ (red), NaGdF.sub.4:Mn.sub.4.sup.+(red), and the like may be used as the fluorescent material. However, the kind of the fluorescent material is not limited thereto.

(108) In other example embodiments, a wavelength conversion material, such as a quantum dot, may be further positioned over the fluorescent layer **160**. The quantum dot may have a core-shell structure using a III-V or II-VI compound semiconductor. For example, the quantum dot may have a core such as CdSe and InP and a shell such as ZnS and ZnSe. In addition, the quantum dot may include a ligand for stabilizing the core and the shell.

(109) A support substrate **170** may be on the wiring structure **140**, and an adhesive layer **172** may be interposed between the support substrate **170** and the wiring structure **140**. In example embodiments, the adhesive layer **172** may include an electrically insulating material, for example, silicon oxide, silicon nitride, a polymer material such as an ultraviolet (UV)-curable material, or resin. In some example embodiments, the adhesive layer **172** may include a eutectic adhesive material, such as AuSn or NiSi. The support substrate **170** may include a sapphire substrate, a glass substrate, a transparent conductive substrate, a silicon substrate, or a silicon carbide substrate, but is not limited thereto.

(110) In general, a light source module including a plurality of light-emitting device chips may be used for an intelligent lighting system (e.g., a head lamp for a vehicle), and each of the light-emitting device chips may be individually controlled to implement various lighting modes depending on surrounding conditions. When a plurality of light-emitting devices arranged in a matrix form are used, light emitted from each of the plurality of light-emitting devices may be absorbed or penetrated into a light-emitting device adjacent thereto. Thus, contrast characteristics of the light source module may be poor. However, according to example embodiments, by forming the partition wall structures WS on the plurality of light-emitting structures **120**, the absorption or penetration of light emitted from one light-emitting cell into an adjacent light-emitting cell may be reduced or prevented.

(111) During a process of forming a passivation layer covering the partition wall structure WS, a thickness of a portion of the passivation layer, which is formed on the top surface of the partition wall structure WS, may be greater than a thickness of a portion of the passivation layer, which is formed on a sidewall of the partition wall structure WS. In this case, the portion of the passivation layer, which is formed on the top surface of the partition wall structure WS, may act as a light guide, and thus, light emitted from one light-emitting cell may be adsorbed or penetrated into an adjacent light-emitting cell. However, according to example embodiments, the first portion **150**P**1** of the passivation structure **150** may be formed to a thickness less than or equal to a thickness of the second portion **150**P**2** thereof. Thus, the absorption or penetration of light emitted from one light-emitting cell into an adjacent light-emitting cell through the first portion **150**P**1** of the passivation structure **150** may be prevented or reduced. Accordingly, contrast characteristics of the light-emitting device **100** may be excellent.

(112) In addition, the fluorescent layer **160** may be firmly fixed in each of the cell spaces PXS by the partition wall structure WS. Even when repetitive vibration and impact are applied to the light-emitting device **100** when the light-emitting device **100** is used as a headlamp for a vehicle, the reliability of the light-emitting device **100** may be improved.
(113) FIGS. **10** and **11** are circuit diagrams for explaining a connection relationship between light-emitting cells included in a light-emitting device **100** according to an example embodiment. FIGS. **10** and **11** are equivalent circuit diagrams of the first cell block BLK**1** and the second cell block BLK**2** of FIG. **4**, according to example embodiments. In FIGS. **10** and **11**, one light-emitting cell may correspond to one diode. The description of the first and second cell blocks BLK**1** and BLK**2** of FIGS. **10** and **11** may be equally applied to the third to ninth cell blocks BLK**3** to BLK**9** of FIG. **4**.
(114) Referring to FIGS. **4** and **10**, a first cell block BLK**1**A may include a plurality of first light-emitting cells **111_1**A, each of which is implemented as an LED, and a second cell block BLK**2**A may include a plurality of second light-emitting cells **111_2**A, each of which is implemented as an LED. The number of first light-emitting cells **111_1**A may be less than the number of second light-emitting cells **111_2**A. The first cell block BLK**1**A and the second cell block BLK**2**A may be electrically insulated from each other, and operations of the first cell block BLK**1**A and the second cell block BLK**2**A may be controlled by different controllers.
(115) In an example embodiment, the light-emitting cells **111** included in each of the first to ninth cell blocks BLK**1**A to BLK**9** may be connected in parallel to other light-emitting cells **111** within the same block. For example, the plurality of first light-emitting cells **111_1**A included in the first cell block BLK**1**A may be connected in parallel to each other, and the plurality of second light-emitting cells **111_2**A included in the second cell block BLK**2**A may be connected in parallel to each other. One end of each of the plurality of first light-emitting cells **111_1**A and the plurality of second light-emitting cells **111_2**A may be respectively connected to different pads, and another end of each of the plurality of first light-emitting cells **111_1**A and the plurality of second light-emitting cells **111_2**A may be connected to a common pad.
(116) Anodes of the plurality of first light-emitting cells **111_1**A may be connected to each other, and anodes of the plurality of second light-emitting cells **111_2**A may be connected to each other. In contrast to the illustration of FIG. **10**, cathodes of the plurality of first light-emitting cells **111_1**A may be connected to each other, and cathodes of the plurality of second light-emitting cells **111_2**A may be connected to each other. The number of first pads PAD**1**A connected to the plurality of first light-emitting cells **111_1**A may be less than twice the number of first light-emitting cells **111_1**A. The number of second pads PAD**2**A connected to the plurality of second light-emitting cells **111_2**A may be less than twice the number of second light-emitting cells **111_2**A.
(117) A first controller may be electrically connected to the plurality of first pads PAD**1**A. The first controller may adjust voltages applied to two different pads, from among the plurality of first pads PAD**1**A, and drive one light-emitting cell of which a cathode and an anode are respectively connected to the two pads. A second controller may adjust voltages applied to two different pads, from among the plurality of second pads PAD**2**A, and drive one light-emitting cell of which a cathode and an anode are respectively connected to the two pads.
(118) Referring to FIG. **11**, a first cell block BLK**1**A′ may further include a plurality of first light-emitting cells **111_1**A and a dummy light-emitting cell **111_1**A′, each of which is implemented as an LED. The dummy light-emitting cell **111_1**A′ may not be connected to a plurality of first pads PAD**1**A and may not substantially operate. In an example embodiment, the number of first light-emitting cells **111_1**A and the dummy light-emitting cell **111_1**A′ included in the first cell block BLK**1**A′ may be equal to the number of second light-emitting cells **111_2**A included in the second cell block BLK**2**.
(119) FIG. **12** is a cross-sectional view of a light-emitting device **300** according to example

embodiments. FIG. **13** is an enlarged view of region CX**4** of FIG. **12**. FIG. **12** is an enlarged cross-sectional view of some components of the light-emitting device **300** adhered onto a printed circuit board **370**, which is taken along line II-II′ of FIG. **7**, according to an example embodiment. In the descriptions of FIGS. **12** and **13**, repeated descriptions of the same reference numerals as given with respect to FIGS. **8** and **9** are omitted.

(120) Referring to FIGS. **12** and **1**, the light-emitting device **300** may further include a buffer structure BS provided between a first conductive semiconductor layer **122** and a partition wall structure WS. The buffer structure BS may include a nucleation layer **310**, a dislocation-removing structure **320**, and a buffer layer **330**, which are sequentially arranged on a bottom surfaces of the partition wall structure WS toward the first conductive semiconductor layer **122**. The dislocation-removing structure **320** may include a first dislocation-removing material layer **322** and a second dislocation-removing material layer **324**.

(121) In example embodiments, the nucleation layer **310** may be a layer for forming nuclei for crystal growth or assisting the wetting of the dislocation-removing structures **320**. For example, the nucleation layer **310** may include aluminum nitride (AlN). The first dislocation-removing material layer **322** may include B.sub.xAl.sub.yInzGa.sub.1-x-y-zN (where 0≤x<1, 0<y<1, 0≤z<1, and 0≤x+y+z<1). In some example embodiments, an aluminum (Al) content of the first dislocation-removing material layer **322** may be about 25 atomic percent (at %) to about 75 at %. The second dislocation-removing material layer **324** may have a different lattice constant from the first dislocation-removing material layer **322** and include, for example, aluminum nitride (AlN). At an interface between the first and second dislocation-removing material layers **322** and **324**, a dislocation may be bent or a dislocation half-loop may be formed due to a difference in lattice constant between the first and second dislocation-removing material layers **322** and **324** to reduce the dislocation. A thickness of the second dislocation-removing material layer **324** may be less than a thickness of the nucleation layer **310**. Thus, tensile stress generated in the second dislocation-removing material layer **324** may be reduced to prevent cracks from occurring.

(122) The buffer layer **330** may reduce differences in lattice constant and coefficient of thermal expansion (CTE) between a layer (e.g., the first conductive semiconductor layer **122**) formed on the buffer structure BS and the second dislocation-removing material layer **324**. For example, the buffer layer **330** may include B.sub.xAl.sub.yIn.sub.zGa.sub.1-x-y-zN (where 0≤x<1, 0<y<1, 0≤z<1, and 0≤x+y+z<1).

(123) According to example embodiments, the buffer structure BS may not only prevent cracks from occurring in a plurality of light-emitting structures **120** but also prevent the propagation of dislocations into the plurality of light-emitting structures **120**. Thus, the crystal quality of the light-emitting structures **120** may be improved.

(124) In example embodiments, a wiring structure **140** may be arranged over a top surface of a support structure **360** with a first adhesive layer **352** therebetween, and a printed circuit board (PCB) **370** may be arranged on a bottom surface of the support structure **360** with a second adhesive layer **354** therebetween. The support structure **360** may include a support substrate **362**, a first insulating layer **364**, and a second insulating layer **366**. The first insulating layer **364** may be in contact with the first adhesive layer **352**, and the second insulating layer **366** may be in contact with the second adhesive layer **354**.

(125) The support substrate **362** may include an insulating substrate or a conductive substrate. The support substrate **362** may have an electric resistance of at least several MΩ, for example, at least 50 MΩ. For example, the support substrate **362** may include doped silicon, undoped silicon, Al.sub.2O.sub.3, tungsten (W), copper (Cu), a bismaleimide triazine (BT) resin, an epoxy resin, polyimide, a liquid crystal (LC) polymer, a copper clad laminate, or a combination thereof. The support substrate **362** may have a thickness of at least 150 μm (e.g., about 200 μm to about 400 μm) in a vertical direction (i.e., a Z direction).

(126) Each of the first insulating layer **364** and the second insulating layer **366** may have an electric

resistance of at least several tens of MΩ, for example, at least 50 MΩ. For instance, each of the first insulating layer **364** and the second insulating layer **366** may include at least one of SiO.sub.2, Si.sub.3N.sub.4, Al.sub.2O.sub.3, HfSiO.sub.4, Y.sub.2O.sub.3, ZrSiO.sub.4, HfO.sub.2, ZrO.sub.2, Ta.sub.2O.sub.5, and La.sub.2O.sub.3.
(127) According to example embodiments, because the support structure **360** has a relatively high electric resistance, the occurrence of failures due to the formation of an undesired conduction path from the wiring structure **140** through the support structure **360** to the PCB **370** in a vertical direction may be prevented.
(128) FIG. **14** is a cross-sectional view of a light-emitting package **1000** including a light-emitting device **100**, according to an example embodiment. In FIG. **14**, the same reference numerals are used to denote the same elements as in FIGS. **1** to **13**.
(129) Referring to FIG. **14**, the light-emitting package **1000** may include the light-emitting device **100** and a semiconductor driving chip **210**, which are mounted on a package substrate **410**. The semiconductor driving chip **210** may be one of the first to ninth controllers **210_1** to **210_9** of FIG. **2**, and the package substrate **410** may be a printed circuit board PCB of FIG. **2**.
(130) A lower insulating layer **412**, an inner conductive pattern layer **413**, and an upper insulating layer **414** may be sequentially stacked on a partial region of a base plate **411**, and at least one driver semiconductor chip **210** may be mounted on a conductive pattern positioned on the upper insulating layer **414**.
(131) An interposer **450** may be on another region of the base plate **411** with an adhesive layer **440** therebetween, and the light-emitting device **100** may be mounted on the interposer **450**. In example embodiments, the interposer **450** may be the same as the support substrate (refer to **170** in FIG. **8**) included in the light-emitting device **100**, but is not limited thereto. The at least one semiconductor driving chip **210** may be electrically connected to a pad PAD of the light-emitting device **100** through a bonding wire **480** connected to a pad **470**. The at least one semiconductor driving chip **210** may individually or entirely drive a plurality of light-emitting cells of the light-emitting device **100**.
(132) The bonding wire **480** may be encapsulated by a molding resin **460**. The molding resin **460** may include, for example, an epoxy molding compound (EMC), but is not specifically limited. The molding resin **460** may partially encapsulate the light-emitting device **100** so as not to interfere with light emitted from the plurality of light-emitting cells of the light-emitting device **100**.
(133) A heat sink **420** may be adhered onto a bottom surface of the base plate **411**, and a TIM layer **430** may be selectively further positioned between the heat sink **420** and base plate **411**.
(134) The light-emitting devices and/or the light source module described with reference to FIGS. **1** to **13** may be mounted alone or in combination in the light-emitting package **1000**.
(135) FIG. **15** is a cross-sectional view of a light-emitting package **1000**A including a light-emitting device **100**A, according to an example embodiment.
(136) Referring to FIG. **15**, a conductive line CL may electrically connect a first contact **134**A of one light-emitting structure (e.g. light-emitting structure **120** of the FIGS. **8**, **9**, **12**) to a second contact **134**B of another light-emitting structure **120** connected in series to a specific light-emitting structure so that a plurality of light-emitting cells included in one cell block are connected in series to each other. Alternatively, in an example embodiment, the conductive line CL may electrically connect the first contact **134**A of one light-emitting structure **120** to the first contact **134**A of another light-emitting structure **120** connected in parallel to the one light-emitting structure **120** so that a plurality of light-emitting cells included in one cell block are connected in parallel to each other.
(137) The conductive line CL may not be formed only in the light-emitting device **100**A. For example, the conductive line CL may include a conductive pattern formed in a package substrate **410**. The conductive line CL may include a conductive pattern formed in an interposer **450**.
(138) In an example embodiment, the light-emitting device **100**A may not include a plurality of

pads, which are formed on an emission surface (e.g., a surface formed in a third direction (Z)). The light-emitting device **100**A may be connected to a semiconductor driving chip **210** through the conductive line CL formed in the package substrate **410**.
(139) FIG. **16** is a schematic perspective view of a lighting apparatus including a light-emitting device, according to an example embodiment.
(140) Referring to FIG. **16**, a headlamp module **2020** may be installed in a headlamp unit **2010** of a vehicle. A side mirror lamp module **2040** may be installed in an outer side mirror unit **2030**, and a tail lamp module **2060** may be installed in a tail lamp unit **2050**. At least one of the headlamp module **2020**, the side mirror lamp module **2040**, and the tail lamp module **2060** may include one of the light source modules **10** and **10**A, which includes at least one of the light-emitting devices **100** and **100**A described above.
(141) FIG. **17** is a schematic perspective view of a flat-panel lighting apparatus **2100** including a light-emitting device, according to an example embodiment.
(142) Referring to FIG. **17**, the flat-panel lighting apparatus **2100** may include a light source module **2110**, a power supply **2120**, and a housing **2130**.
(143) The light source module **2110** may include a light-emitting device array as a light source. The light source module **2110** may be one of the light source modules **10** and **10**A, which includes, as a light source, at least one of the light-emitting devices **100** and **100**A described above. The light source module **2110** may have a flat shape as a whole.
(144) The power supply **2120** may be configured to supply power to the light source module **2110**. The housing **2130** may form an accommodation space for accommodating the light source module **2110** and the power supply **2120**. The housing **2130** may be formed to have a hexahedral shape with one opened side, but is not limited thereto. The light source module **2110** may be positioned to emit light toward the opened side of the housing **2130**.
(145) FIG. **18** is an exploded perspective view of a lighting apparatus **2200** including a light-emitting device, according to an example embodiment.
(146) Referring to FIG. **18**, the lighting apparatus **2200** may include a socket **2210**, a power supply **2220**, a heat sink **2230**, a light source module **2240**, and an optical unit **2250**.
(147) The socket **2210** may be configured to be replaceable with an existing lighting apparatus. Power may be supplied to the lighting apparatus **2200** through the socket **2210**. The power supply **2220** may be dissembled into a first power supply **2221** and a second power supply **2222**. The heat sink **2230** may include an internal heat sink **2231** and an external heat sink **2232**. The internal heat sink **2231** may be directly connected to the light source module **2240** and/or the power supply **2220** and transmit heat to the external heat sink **2232** through the light source module **2240** and/or the power supply **2220**. The optical unit **2250** may include an internal optical unit and an external optical unit. The optical unit **2250** may be configured to uniformly disperse light emitted by the light source module **2240**.
(148) The light source module **2240** may receive power from the power supply **2220** and emit light to the optical unit **2250**. The light source module **2240** may include at least one light-emitting element package **2241**, a circuit board **2242**, and a controller **2243**. The controller **2243** may store driving information of the light-emitting device packages **2241**. The light-emitting device package **2241** may include at least one of the light-emitting devices **100** and **100**A described above, and the light source module **2240** may be one of the light source modules **10** and **10**A described above.
(149) While example embodiments have been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A light source module comprising: a printed circuit board; a light-emitting device mounted on the printed circuit board, the light-emitting device comprising a plurality of cell blocks, each of which comprises a plurality of light-emitting diodes (LEDs); and a plurality of controllers mounted on the printed circuit board, each of which is configured to drive a cell block corresponding thereto, from among the plurality of cell blocks, wherein the plurality of cell blocks are electrically isolated from each other, wherein the plurality of cell blocks comprise a first cell block and a second cell block, wherein a number of first LEDs included in the first cell block is less than a number of second LEDs included in the second cell block, wherein the plurality of cell blocks and the plurality of controllers are provided on a same side of the printed circuit board, wherein the first LEDs are connected in series to each other and are configured to emit light of a same color, wherein the second LEDs are connected in series to each other and are configured to emit light of a same color, wherein the plurality of controllers comprise a first controller connected to a plurality of first channels and a second controller connected to a plurality of second channels which are different from the plurality of first channels, wherein one of the plurality of first channels is connected between two of the first LEDs, wherein one of the plurality of second channels is connected between two of the second LEDs, wherein the first controller is configured to drive the first cell block through the plurality of first channels, and wherein the second controller is configured to drive the second cell block through the plurality of second channels.
2. The light source module of claim 1, wherein a number of the plurality of first channels connected to the first controller is less than a number of the plurality of second channels connected to the second controller.
3. The light source module of claim 1, wherein the light-emitting device is disposed in a central area of the printed circuit board between the first controller and the second controller, and wherein the plurality of controllers are disposed in a peripheral area of the printed circuit board surrounding the central area.
4. The light source module of claim 1, wherein the first LEDs are electrically connected to each other in series through conductive lines formed on the printed circuit board.
5. The light source module of claim 1, wherein the plurality of cell blocks are arranged in a first row and a second row, wherein the first row and the second row extend in parallel between the first controller and the second controller, and wherein the first cell block is arranged on an end of the first row.
6. The light source module of claim 1, wherein the plurality of cell blocks are arranged in a first row and a second row, wherein the first row and the second row extend in parallel between the first controller and the second controller, and wherein the first cell block is in a center of the second row.
7. The light source module of claim 6, wherein the first cell block comprises a smallest number of LEDs from among the plurality of cell blocks.
8. The light source module of claim 1, wherein the one of the plurality of first channels is connected between an anode of one of the first LEDs and a cathode of another one of the first LEDs.
9. The light-emitting device of claim 1, wherein the LEDs in each of the plurality of cell blocks are arranged in a matrix.
10. The light source module of claim 9, wherein the first cell block is adjacent the second cell block.
11. A light-emitting device comprising: a plurality of cell blocks electrically isolated from each other, each of which comprises a plurality of light-emitting diodes (LEDs) and is provided on a printed circuit board; a plurality of controllers, each of which is configured to drive a cell block corresponding thereto, from among the plurality of cell blocks, and is provided on the printed circuit board; and a plurality of pads configured to electrically connect the plurality of cell blocks

to an external device, wherein the plurality of cell blocks comprise a first cell block and a second cell block, wherein a number of first LEDs included in the first cell block is less than a number of second LEDs included in the second cell block, wherein the first LEDs are connected in series to each other and are configured to emit light of a first color, wherein the second LEDs are connected in series to each other and are configured to emit light of a second color, wherein the plurality of controllers comprise a first controller connected to a plurality of first channels and a second controller connected to a plurality of second channels which are different from the plurality of first channels, wherein one of the plurality of first channels is connected between two of the first LEDs, wherein one of the plurality of second channels is connected between two of the second LEDs, wherein the first controller is configured to drive the first cell block through the plurality of first channels, wherein the second controller is configured to drive the second cell block through the plurality of second channels, and wherein the plurality of cell blocks and the plurality of pads are provided on a same side of the printed circuit board.
12. The light-emitting device of claim 11, wherein the first LEDs further comprise a dummy LED electrically isolated from the plurality of pads.
13. The light-emitting device of claim 11, wherein the plurality of cell blocks are arranged in a first row and a second row, wherein the first row and the second row extend in parallel between the first controller and the second controller, and wherein a total number of LEDs included in cell blocks arranged in the first row is less than a total number of LEDs included in cell blocks arranged in the second row.
14. The light-emitting device of claim 11, wherein the plurality of cell blocks are arranged in a first row and a second row, wherein the first row and the second row extend in parallel between the first controller and the second controller, and wherein the first cell block is arranged on one end of the first row.
15. The light-emitting device of claim 11, wherein the plurality of cell blocks are arranged in a first row and a second row, wherein the first row and the second row extend in parallel between the first controller and the second controller, and wherein the first cell block is in a center of the second row.
16. The light-emitting device of claim 11, wherein the plurality of cell blocks are disposed in a central area of the light-emitting device between the first controller and the second controller, and wherein the plurality of pads are disposed in a peripheral area of the light-emitting device surrounding the central area.
17. A light source module comprising: a printed circuit board; a light-emitting device mounted on the printed circuit board, the light-emitting device comprising a plurality of cell blocks, each of which comprises a plurality of light-emitting diodes (LEDs); and a plurality of controllers mounted on the printed circuit board, each of which is configured to drive a cell block corresponding thereto, from among the plurality of cell blocks, wherein the plurality of cell blocks and the plurality of controllers are provided on a same side of the printed circuit board, wherein the plurality of cell blocks are electrically isolated from each other, wherein the plurality of cell blocks comprise a first cell block and a second cell block, wherein the plurality of cell blocks are arranged in a first row and a second row, and a number of LEDs included in cell blocks arranged in the first row is less than a number of LEDs included in cell blocks arranged in the second row, wherein first LEDs provided in the first cell block are connected in series to each other and are configured to emit light, wherein second LEDs provided in the second cell block are connected in series to each other and are configured to emit light, wherein the plurality of controllers comprise a first controller connected to a plurality of first channels and a second controller connected to a plurality of second channels which are different from the plurality of first channels, wherein the first controller is configured to drive the first cell block through the plurality of first channels and one of the plurality of first channels is connected between two of the first LEDs, and wherein the second controller is configured to drive the second cell block through the plurality of second channels and one of the

plurality of second channels is connected between two of the second LEDs.

18. The light source module of claim 17, wherein two cell blocks arranged on two ends of the first row comprise a smaller number of LEDs than other cell blocks positioned adjacent thereto.

19. The light source module of claim 17, wherein a cell block arranged in a center of the second row comprises a smaller number of LEDs than other cell blocks positioned adjacent thereto.

20. The light source module of claim 17, wherein the plurality of LEDs included in one cell block, from among the plurality of cell blocks, are electrically connected to each other in series through conductive lines formed on the printed circuit board.